

Unless given specifically

Take -- $V_{DD} = V_{CC} = 3.3V$, $L_{min} = 1\mu m$ for $0.5\mu m$ technology.

For NMOS device $\mu_n C_{ox} = 140 \mu A/V^2$, $V_T = 0.7 V$, $\lambda = 0.1 V^{-1}$, $C_{ox} = 0.38 pF/\mu m^2$

For PMOS device $\mu_p C_{ox} = 40 \mu A/V^2$, $V_T = -0.7 V$, $\lambda = 0.1 V^{-1}$, $C_{ox} = 0.38 pF/\mu m^2$

For NPN/ PNP device $\mu_p \text{ Cox} = 40 \text{ } \mu\text{A/V}^2$, $V_T = -0.7 \text{ V}$, $\lambda = 0.1 \text{ V}^{-1}$, $\text{Cox} = 0.38 \text{ pF}/\mu\text{m}^2$, $\beta = 50$, $V_{\text{CESAT}} = 0.2 \text{ V}$, $V_A = 100 \text{ V}$, $KT/q = 25 \text{ mV}$ (at room temp.), $I_s = 10^{-14} \text{ A}$

NOTE:

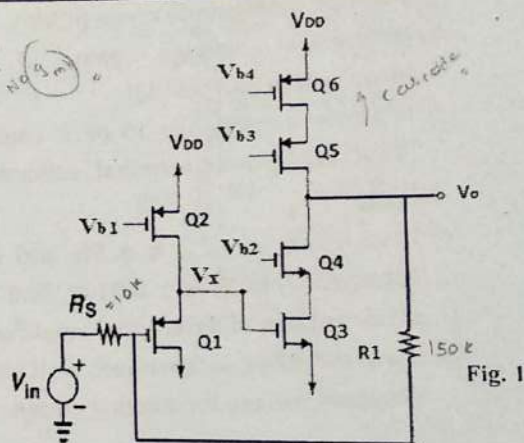
If not specified in question,-----

- Ignore γ , λ in drain current equation. Assume matched components wherever required
- Bulk of nmos connected to ground and bulk of pmos connected to Vdd. Specify your assumptions. Justify your answers.
- Unless specified, assume all MOSFET/BJT are biased in active region
- Label your sketches properly. All symbols have usual meaning
- **Do all parts of the same question together**

Q 1. For the amplifier circuit shown in Fig. 1, $g_{mn} = 2 \text{ mA/V}$, $g_{mp} = 1 \text{ mA/V}$, $r_o = 100 \text{ K}$, $\chi = 0$ (for all the transistors) and resistor values $R_1 = 150 \text{ K}$. If amplifier is connected to a source having $R_s = 10 \text{ K}$ then

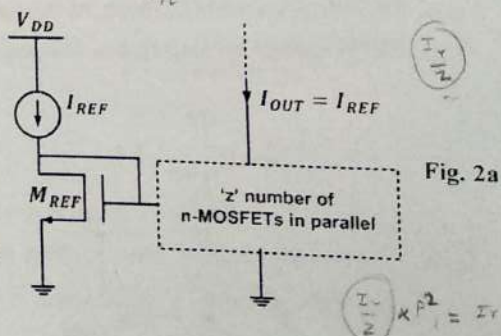
- Identify the type of sampling, type of mixing and type of feedback topology (amplifier).
- Calculate the feedback factor β .
- Calculate the appropriate open loop gain [for the feedback topology of part (a)] including the loading effect of feedback using approximated intuitive techniques.
- Hence, Calculate the corresponding closed loop gain.
- Now, calculate the overall voltage gain (V_o/V_{in}) with feedback.

(22 M)

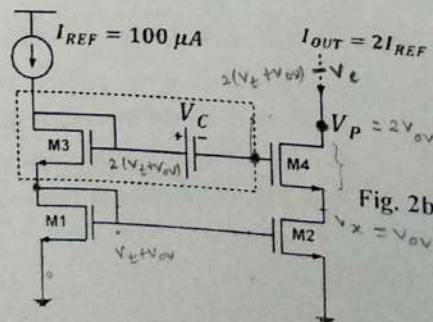


Q 2. (a) Fig. 2a shows a basic current mirror where $I_{OUT} = I_{REF}$.

The transistor M_{REF} is connected to 'z' number of n-MOSFETs in parallel (marked as a dotted box) where each n-MOSFET has aspect ratio of $p^2 \left(\frac{W}{L} \right)$. If M_{REF} has $\left(\frac{W}{L} \right)_{M_{REF}} = 343 \left(\frac{W}{L} \right)$, find the values of 'z' and 'p' considering that both are whole numbers. Neglect channel length modulation in your calculations.



Q2. (b) For the circuit shown in Fig. 2b. If the overdrive voltages (V_{OV}) of all the transistors are same, find the value of V_C so that the minimum value of V_P ($V_{P,min}$) required to make all the transistors in saturation is $2V_{OV}$.



$$\frac{1_{\text{TOT}}}{2 \rho^2 \omega_L} = \frac{1_{\text{TOT}}}{2 \rho^2 \omega_L}$$

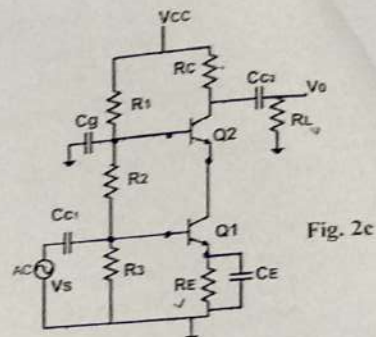
$P^2Z = 343$

$$v_c = v_b \quad v_g - (v_t + v_{ov}) - v_t = v_{ov}$$

$$V_{ac} - V_t = V_o$$

$$V_g - V_x - V_t = 2V_{ov} - V_{th}$$

Q2 (c) Consider the circuit shown in Fig. 2c, Let $\beta = 100$, $V_{BE(ON)} = 0.7$ V, and $V_A = \infty$ for each transistor. Assume $V_{CC} = 3.3$ V, $R_L = 2$ k Ω , and $R_E = 0.5$ k Ω .



- Identify the circuit topology.
- Find R_C , R_1 , R_2 , R_3 such that $I_{CQ2} = 0.5$ mA and $V_{CE1} = V_{CE2} = 0.9$ V. Let $R_1 + R_2 + R_3 = 100$ k Ω . (Hint: Neglect the DC base currents and assume $I_C = I_E$ in both Q_1 and Q_2 .)
- Determine the value of small-signal hybrid- π parameters for each transistor.
- Neatly sketch and label the small-signal equivalent circuit at high frequency.

(22 M)

Q3 Fig.3 shows low-voltage, folded-current mirror single ended differential amplifier. Assume all the transistors has same aspect ratio ($W/L = 5/1$), $I_{ref} = 50$ μ A and $V_{DD} = 3.3$ V.

Find the values of following--

- Input common mode range (ICMR).
- Differential voltage gain ($A_v = v_{out}/(v_{in1} - v_{in2})$) intuitively in decibel (dB).
- If a load capacitance 10 pF is connected between node 'Vout' and ground terminal, estimate the -3 dB frequency ' ω_{-3dB} '?
- If variation in $\mu_n C_{ox}$ is $\pm 5\%$ and variation in threshold voltage of nMOS is ± 0.01 V, find out worst case input offset voltage of differential amplifier.
- If a low voltage design needs an ICMR of at least 0.5 V, than what could be the least possible value of supply voltage for existing design.

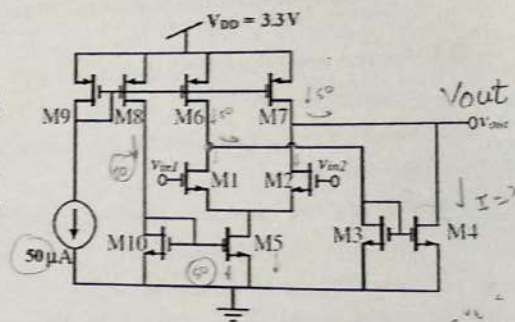


Fig. 3

Q4 (a) Consider Fig. Q4 [(a), (b), and (c)]. Current sources I_{b1} and I_{b2} are nearly ideal sources. In each circuit, (M1, M2), (M3, M4), and (Q1, Q2) are matched transistor. Assume all transistors are operating in active region. Assume a total load capacitor of C_L at node $vo1$.

(21 M)

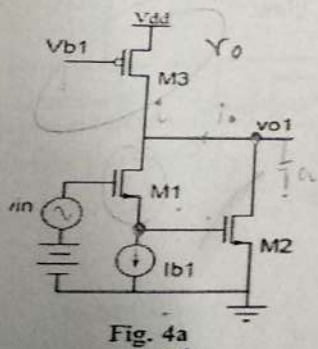


Fig. 4a

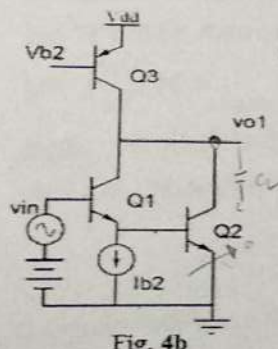


Fig. 4b

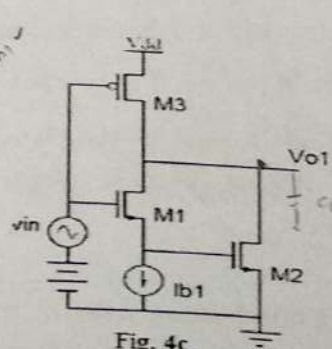


Fig. 4c

Intuitively, derive the expression for low frequency voltage gain $A_v = [vo1/vin]$ using G_m and R_{out} of each circuit. Also derive the expression for magnitude of pole frequency ' ω_{out} ' at node $vo1$ intuitively for each. Hence fill the entries in Table--1 given below. {Take that all transistors have identical g_m and r_o .}

$$= \frac{1}{R_{out} \times C_L}$$

Circuit	TABLE-I		
	G_m	R_{out}	Output pole frequency ω_{out}
Fig. Q1a			
Fig. Q1b			
Fig. Q1c			

Q4 (b) Consider the circuits shown in Fig. 4[(e), (f) & (g)]. (M1, M2), (M3, M4, M5, M6) are matched transistor. (Q1, Q2, Q3, Q4) and (Q5, Q6) are matched transistor. Assume all transistors are operating in active region. I_{ss} is near ideal current source with impedance R_{ss} , and $V_{id} = v_{in1} - v_{in2}$.

Given $I_{ss} = 100\mu A$, $V_{ov} = 0.2V$, $C_L = 5pF$, capacitance $C_{\pi} = C_{gs} = 1pF = 100C_{gd} = 100C_{\mu}$. Neglect other parasitic capacitances. For calculations, assume p node at ac ground and use intuitive method primarily.

Here for BJT, take $\beta = 20$

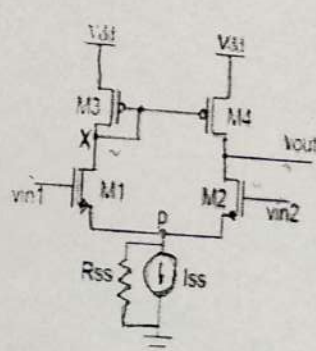


Fig. 4e

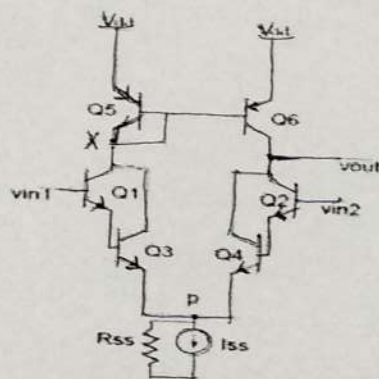


Fig. 4f

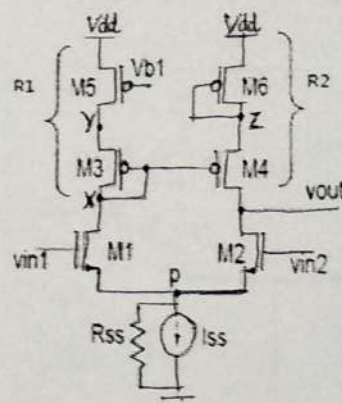


Fig. 4g

For, both, (Fig. 4e, and Fig. 4f)---

- Find the expression and values of all poles and zero frequencies intuitively. Sketch a neat and labeled (not to the scale but qualitatively correct) Bode magnitude plot for both circuits on same axis. Clearly mark low frequency voltage gain ($= v_{out}/v_{in}$) (in dB), and 3-dB bandwidth (in rad./sec) on the plot for each circuit clearly.

In Fig. 4g, -----

- Is the load $R1 = R2$? Justify by writing the complete expressions for $R1$, and $R2$.
- Find the expression and values of low frequency voltage gain $[= v_{out}/v_{id}]$ (in dB). Here, also, find the expression for pole frequency due to mirror node 'x' assuming r_o very large for (M3, M4, M5, M6).
- Assume transistor M6 is removed i.e node z is connected to V_{DD} . Calculate the value of α in $\left[\left(\frac{W}{L}\right)_4 = \alpha \left(\frac{W}{L}\right)_3\right]$ for each arm to carry equal bias current of $(I_{ss}/2)$. Hence calculate the new value for low frequency voltage gain $[= v_{out}/v_{id}]$, and output voltage swing.
- Derive the new expression for low frequency voltage gain $[= v_{out}/v_{id}]$ if node z is connected to V_{b1} .